Hongli Shen

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Education

Southern University of Science and Technology (SUSTech) Sep. 2023 – Present
Bachelor of Engineering in Computer Science and Technology *Shenzhen, China*
GPA: 3.89/4.00 **Average Score:** 93.06/100 **Rank:** 12/152
Relevant Coursework: Algorithm Design and Analysis (99), Introduction to Mathematical Logic (H) (98), Advanced Programming (97), Principles of Database Systems (95), Natural Language Processing (94), Deep Learning (93), Discrete Mathematics (93), Object-Oriented Analysis and Design (92), Linear Algebra (90), Probability and Statistics for Engineering (90), Artificial Intelligence (88).

National University of Singapore (NUS) May 2025 – July 2025
School of Computing Summer Workshop *Singapore*
Grade: A-

Research Interests

Large Language Models, Large Reasoning Models, Multimodal Large Language Models, AI Agents.

Publications

1. [Preprint] **Hongli Shen**[*], Shaopeng Fu[*], Qinbo Zhang, Jian Li, and Di Wang. "Dual-Adversarial Safety Alignment: Cultivating Intrinsic Threat Comprehension in LRMs." *arXiv preprint arXiv:2608.09542*, 2026. **Under review at ICLR 2027.** [Code]

2. [Paper] Yan Hu, Mingdao Gong, Zhongxi Qiu, Jiabao Liu, **Hongli Shen**, Mingzhen Yuan, Xiaoqing Zhang, Heng Li, Hai Lu, and Jiang Liu. "COph100: A Comprehensive Fundus Image Registration Dataset from Infants Constituting the "RIDIRP" Database." *Scientific Data*, 12, 99, 2025.

[*]Equal contribution.

Research Experience

Chain-of-Thought Reconstruction for Large Reasoning Models June 2026 – Present
PRADA Lab, KAUST | Advisor: Prof. Di Wang *Thuwal, Saudi Arabia*

- Investigating chain-of-thought reconstruction in large reasoning models, with a focus on recovering and understanding reasoning trajectories to improve model interpretability and reliability.
- First-authored the resulting paper, currently under review at **ICLR 2027**.

Safety Alignment for Large Reasoning Models Feb. 2026 – Aug. 2026
PRADA Lab, KAUST | Advisor: Prof. Di Wang *Thuwal, Saudi Arabia*

- Proposed **AdvSafe**, a dual-adversarial safety alignment framework that trains large reasoning models to internalize harmful intent and adversarial mechanisms rather than memorizing surface-level refusal patterns.
- Developed an **agentic jailbreak synthesis pipeline** and a teacher-guided adversarial extraction process to construct structured safety reasoning traces, and fine-tuned student LRMs using LoRA-based supervised fine-tuning.
- Improved jailbreak robustness and out-of-distribution safety generalization while preserving general reasoning utility.
- First-authored the resulting paper, *Dual-Adversarial Safety Alignment: Cultivating Intrinsic Threat Comprehension in LRMs*, currently under review at **ICLR 2027**.

Ophthalmic Image Registration Aug. 2024 – Nov. 2024
iMed Lab, SUSTech | Advisor: Prof. Jiang Liu *Shenzhen, China*

- Conducted research on fundus image processing and retinal image registration, focusing on challenging infant ophthalmic images with substantial variations in image quality and acquisition conditions.
- Contributed to the construction and evaluation of **COph100**, a comprehensive infant fundus image registration benchmark consisting of 100 eyes and 491 image pairs from the RIDIRP database.
- Contributed to dataset annotation, preprocessing, and benchmark evaluation for retinal image registration; the resulting work was published in **Scientific Data** in 2025.

Ongoing Research Projects

Environment Scaling for LLM Agents June 2026 – Present
RUC | Advisor: Shaolei Zhang *Beijing, China*

- Developing scalable interactive workspace environments for training and evaluating autonomous LLM agents on diverse and long-horizon tasks.
- Investigating how scaling the diversity and complexity of interactive environments can improve agent capabilities and generalization.

Honors & Awards

SUSTech Merit Student Scholarship (First Class, Top 2%) Nov. 2024, Nov. 2025
Star of Sports, Shuli College Dec. 2024, Dec. 2025
Provincial Third Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) Sep. 2024
SUSTech Outstanding Freshman Scholarship Dec. 2023

Leadership & Activities

Captain, Shuli College Football Team
Member, SUSTech Varsity Football Team

Skills

Programming	Python, C/C++, Java, SQL, JavaScript
Machine Learning & LLMs	PyTorch, LoRA, Supervised Fine-Tuning, LLM and Agent Development
Development Tools	Git, FastAPI, PostgreSQL, React
Hardware	RISC-V, Verilog, FPGA
Languages	Chinese (Native), English (Fluent)
English Proficiency	TOEFL: 5.5 (Reading: 4.5, Listening: 5.5, Speaking: 6.0, Writing: 5.5)